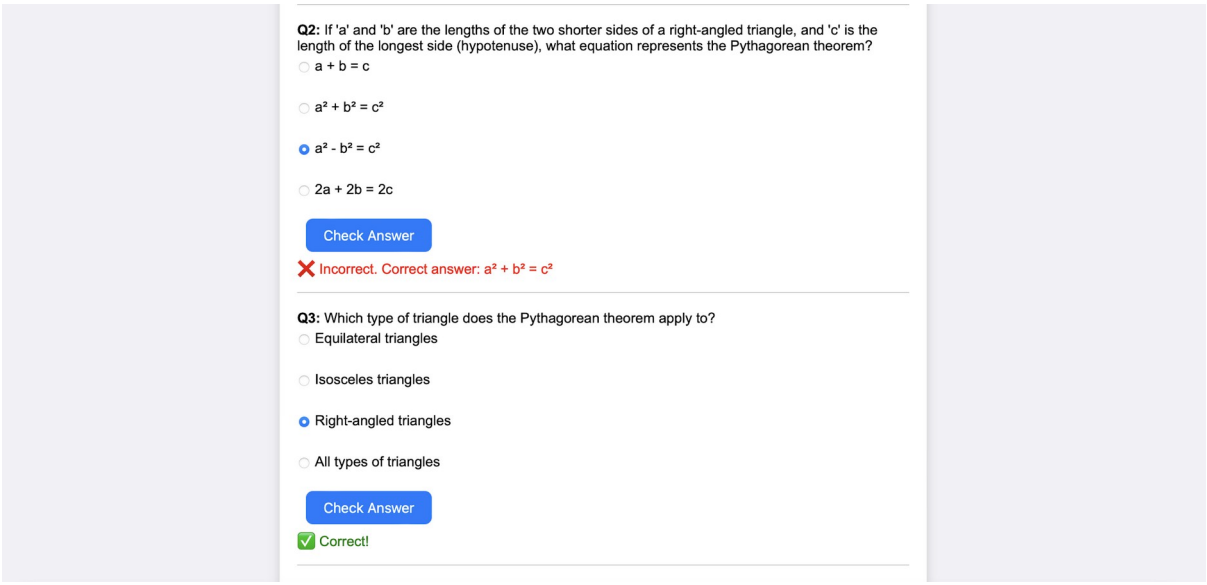
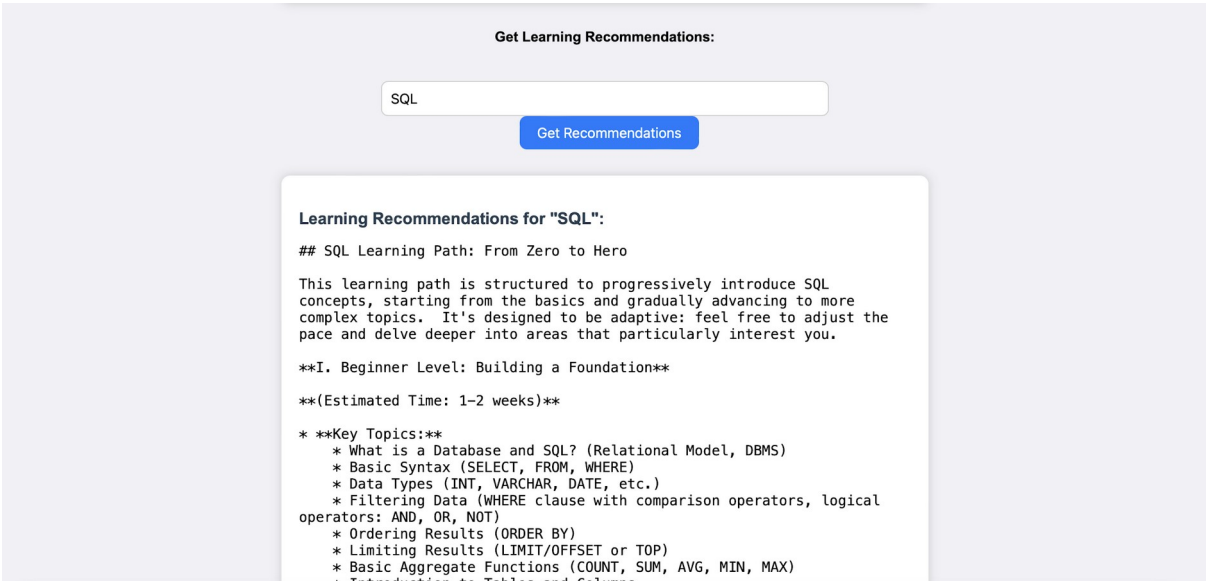


e. Guiding with learning recommendations from beginners level to advanced level



```
complex topics.  It's designed to be adaptive: feel free to adjust the
pace and delve deeper into areas that particularly interest you.

**I. Beginner Level: Building a Foundation**

**(Estimated Time: 1-2 weeks)**

* **Key Topics:**
    * What is a Database and SQL? (Relational Model, DBMS)
    * Basic Syntax (SELECT, FROM, WHERE)
    * Data Types (INT, VARCHAR, DATE, etc.)
    * Filtering Data (WHERE clause with comparison operators, logical
operators: AND, OR, NOT)
    * Ordering Results (ORDER BY)
    * Limiting Results (LIMIT/OFFSET or TOP)
    * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
    * Introduction to Tables and Columns

* **Resources:**
    * **Interactive Tutorials:**
        * SQLZoo: Excellent for hands-on practice with different SQL
flavors.
        * Khan Academy's SQL course: Offers a good introductory
overview.
        * Codecademy's Learn SQL: Interactive lessons with practical
exercises.
    * **Videos:**
        * FreeCodeCamp's SQL Tutorial: Comprehensive and beginner-
friendly.


**II. Intermediate Level: Working with Multiple Tables**

**(Estimated Time: 2-3 weeks)**
```


```
**II. Intermediate Level: Working with Multiple Tables**

**(Estimated Time: 2-3 weeks)**
```


```
* **Key Topics:**
    * Joining Tables (INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER
JOIN)
    * Subqueries (Nested queries)
    * Grouping Data (GROUP BY)
    * Filtering Groups (HAVING)
    * Set Operations (UNION, INTERSECT, EXCEPT)
    * Working with Strings (string concatenation, substrings, LIKE
operator)
    * Working with Dates and Times (date formatting, date arithmetic)
    * Views (Creating and using views)

* **Resources:**
    * **Books:**
        * "SQL Queries for Mere Mortals" by Michael J. Hernandez: A
great resource for understanding practical SQL querying.
    * **Online Documentation:** Refer to the documentation for your
specific SQL database (e.g., MySQL, PostgreSQL, SQL Server).
    * **Practice Platforms:** LeetCode, HackerRank, and similar
platforms offer SQL challenges to reinforce your skills.

**III. Advanced Level: Mastering Database Management**

**(Estimated Time: 3-4 weeks and beyond)**

* **Key Topics:**
    * Stored Procedures and Functions
    * Triggers
    * Indexes and Performance Tuning
    * Transactions and Concurrency Control
    * Database Design (Normalization)
```